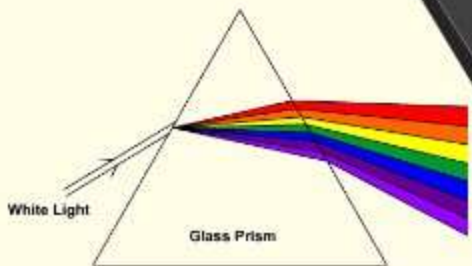


PHYSICS

THE CONCEPTS OF PHYSICAL QUANTITIES

MODULE - 1



LOOK AT THE FOLLOWING...



TIME



LENGTH



MASS



LOVE



DILIGENCE

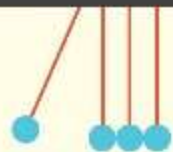


THOUGHTS

CAN YOU FIND THE DIFFERENCE?

CAN BE MEASURED

CANNOT BE MEASURED



PHYSICAL QUANTITY



TWO APPLES



ONE COT



2KG CAKE



1 LITRE MILK

DEFINITION

Physical Quantities are those which can be measured or quantified.





PHYSICAL QUANTITIES - FEATURES

MAGNITUDE

UNIT

DIRECTION



PHYSICAL QUANTITIES - FEATURES

1

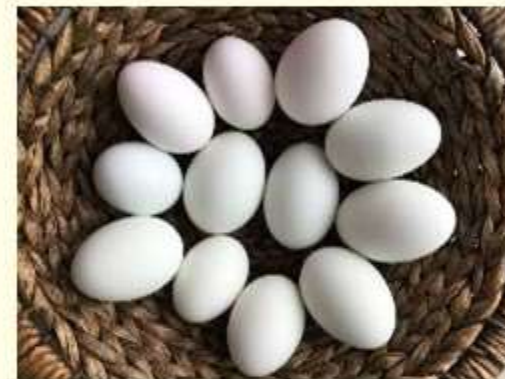
MAGNITUDE



4 MANGOES



3 SHIRTS



12 EGGS

What is the significance of the numbers 4, 3 and 12?

MAGNITUDE

Includes the numeric value of the quantity

PHYSICAL QUANTITIES - FEATURES

2

UNIT



5 kg Rice



10 cm

7 cm



1 Tera Byte

What is the significance of the *kg*, *cm* and *Tera*?

UNIT

A definite standard by which we can quantify magnitude.

PHYSICAL QUANTITIES - FEATURES

3

DIRECTION



CAR IS MOVING IN A
DIRECTION



ROCKET
MOVING UP



BALL FALLING
DOWN

DIRECTION

This defines where or in which way the quantity is moving



TYPES OF PHYSICAL QUANTITY

PHYSICAL QUANTITY

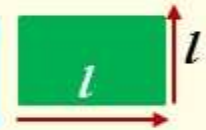
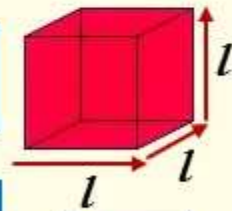
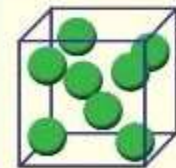
BASE QUANTITY

LENGTH

LENGTH

MASS

DERIVED QUANTITY

AREA ($l \times l$)VOLUME ($l \times l \times l$)DENSITY
(Mass / Volume)

BASE QUANTITIES

QUANTITY	SI UNIT	SYMBOL
Length	Meter	m
Mass	Kilogram	kg
Time	Second	s
Electric current	Ampere	A
Temperature	Kelvin	K
Amount of Substance	Mole	mol
Luminous Intensity	Candela	cd

**FUNDAMENTAL
QUANTITIES**

PHYSICAL QUANTITY ... HOW IT CAN BE EXPRESSED?

UNIT

CGS

Centimeter
Gram
Second

FPS

Foot
Pound
Second

MKS

Meter
kilogram
Second

SI Units



PHYSICAL QUANTITY - EXPRESSION

SI SYSTEM - INTERNATIONAL SYSTEM OF UNITS

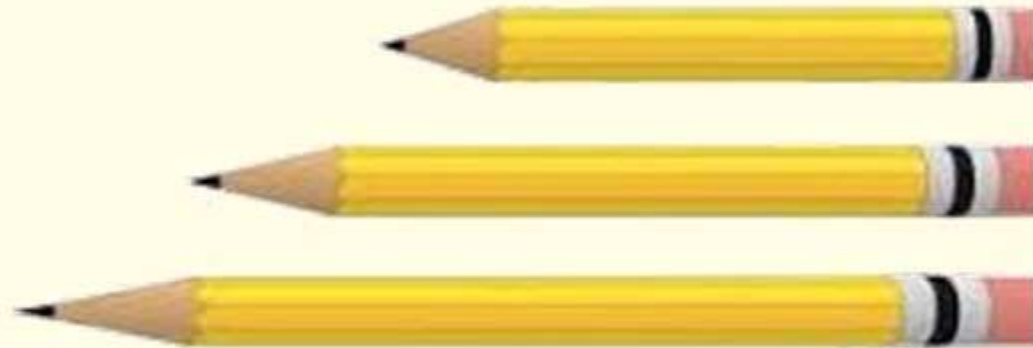


- 1 In 1960 the International System of Units was proposed as a replacement for the Metric System.
- 2 It is based on MKS system i.e. meter kilogram and second.
- 3 SI system has seven base units.

The International Bureau of Weights and Measure located in Sevres France is the international Organization to maintain the SI units.

BASE QUANTITIES

1 LENGTH



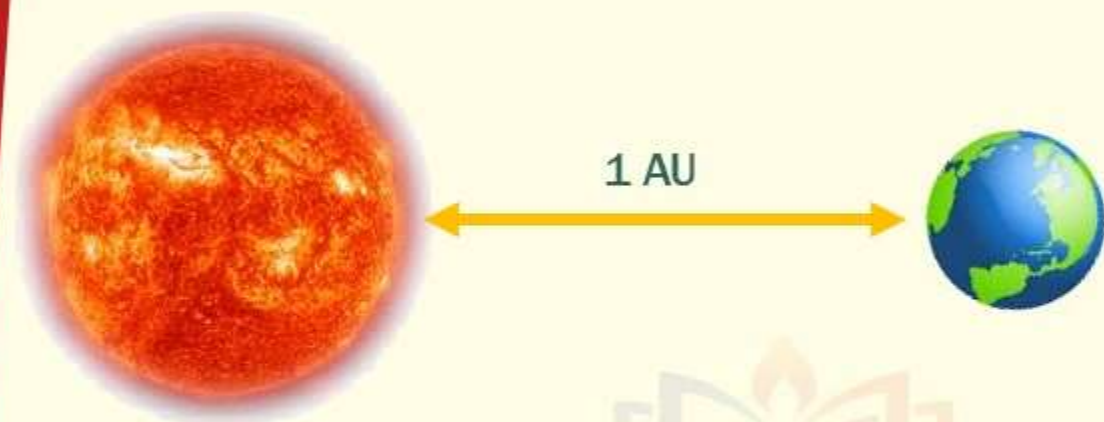
LENGTH

- Measurement of the extent of something
- It is expressed in *METER*



BIGGER INTERVALS OF DISTANCE

PARSEC, LIGHT YEAR & ASTRONOMICAL UNIT



Parsec > Light-year > AU

- **Light year** is used to measure astronomical distance which is the distance taken by light to travel for a year.
- 1 **Astronomical Unit** is defined as **150 million kilometres** which is the average distance between Sun and Earth.
- AU is useful to measure the average distance of Sun and the planets in or around the Solar System.
- But outside the Solar System a larger unit of measurement is used called as **Parsec**
- 1 parsec = 2,06,265 AU.
- 1 parsec = 3.26 light year
- 1 light year = 63,241 AU

SEVEN BASE QUANTITIES

2

TIME



- Measurement of duration between two events.
- *Time is expressed in seconds*
- More smaller units than seconds :
 - Microseconds
 - Nanoseconds
 - Picoseconds
 - Femtoseconds



SEVEN BASE QUANTITIES

3

MASS

- *Measurement of the amount of matter the body contains.*
- *Mass is expressed in Kilogram.*
- Mass is measured accurately using a balance like physical balance or digital balance.



DO YOU THINK MASS AND WEIGHT IS SAME ?

EARTH



Mass = 10 kg
Weigh scales = 10 kg
Weight = 98 N

MOON



Mass = 10 kg
Weigh scales = 1.6 kg
Weight = 16 N

SPACE



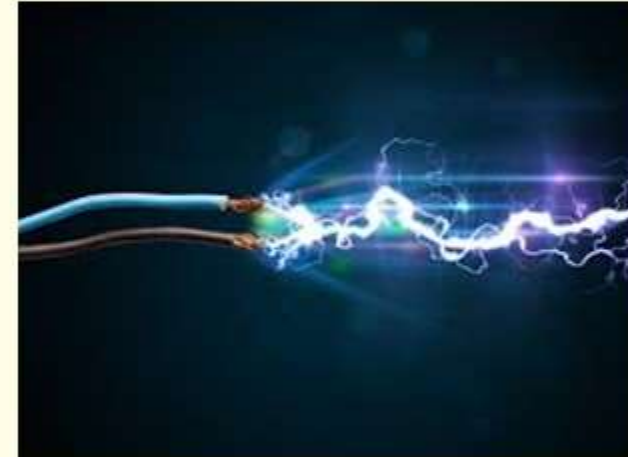
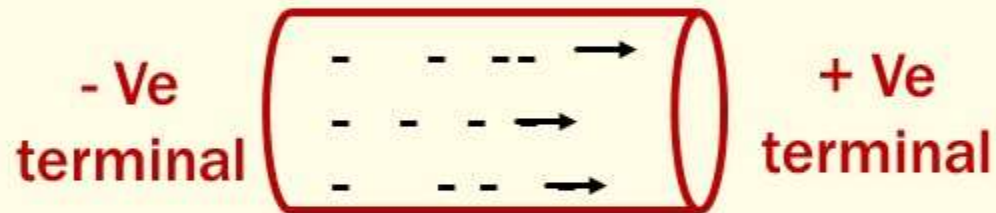
Mass = 10 kg
Weigh scales = 0 kg
Weight = 0 N

Weighing scale measures weight but brings it down to the mass



SEVEN BASE QUANTITIES

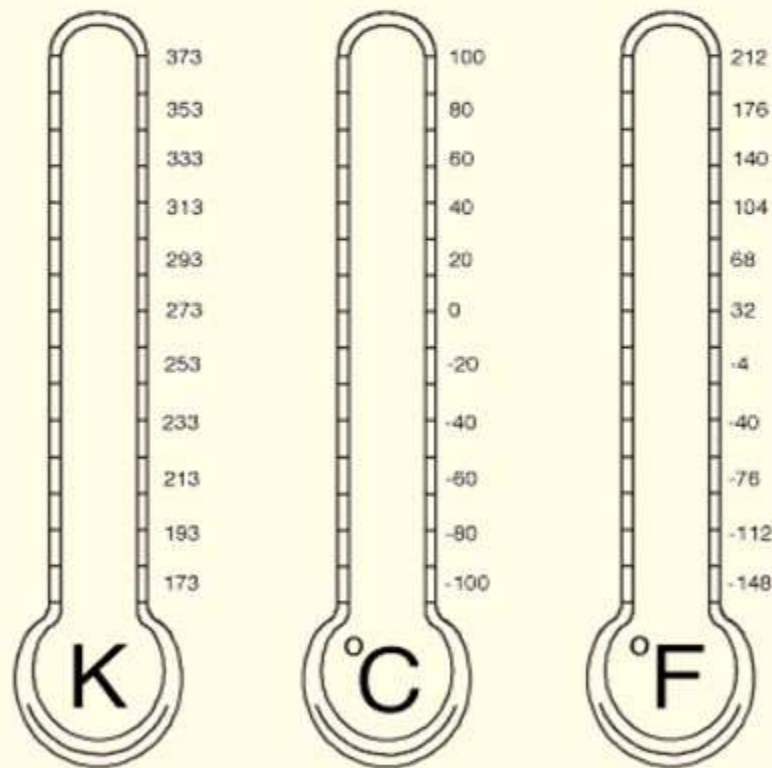
4 | ELECTRIC CURRENT



- Measurement of flow of electric charge.
- Electric current is expressed in Ampere

SEVEN BASE QUANTITIES

5 TEMPERATURE



Measurement of the intensity of the heat present in the substance.

Temperature is expressed in Kelvin

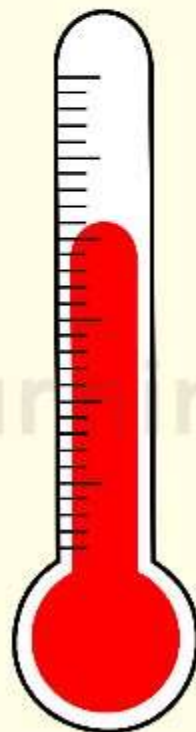
- Other than Kelvin Two different scales of temperature are used
- **Celsius Scale** which we measure general weather
- **Fahrenheit Scale** which we generally measure body Temperature

DO YOU KNOW!!

HEAT AND TEMPERATURE ARE ACTUALLY DIFFERENT !

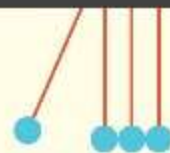
HEAT

- This is a **qualitative measurement**
- Finds whether a substance is hot or cold
- This is a **form of energy**
- This is a **derived unit**
- This is measured in **Joules**.



TEMPERATURE

- This is a **quantitative measurement**
- Finds the amount of hot or cold content in the substance
- This is **not a form of energy**
- This is a **base unit**
- This is measured in **Kelvin**



SEVEN BASE QUANTITIES

6 AMOUNT OF SUBSTANCE



- Amount of Substance measures the particles present inside or the mass of one definitive quantity of substance.
- *Expressed in moles*
- $1 \text{ mole} = 6.022 \times 10^{23} \text{ atoms}$

1 MOLE CAN BE DIFFERENT IN MASS IN DIFFERENT SUBSTANCES!

Mole is a amount of substance that contains the same number of particles (atoms, ions, molecules etc.) as there are atoms in 12 g of carbon-12

TABLE SALT



1 MOLE = 58.4 g

NEELA THOTHA



1 MOLE = 159.6 g

GLUCOSE



1 MOLE = 180.2 g



SEVEN BASE QUANTITIES

7 INTENSITY OF LIGHT

40W
Candle



- Measurement of the amount of light energy released from an unit area
- *Expressed in candela*
- A candle emits light intensity of about 1 candela.
- Whereas a 25W florescent bulb(commonly used) produces about 135 candela



PHYSICAL QUANTITY - COMPONENTS

PHYSICAL QUANTITY

SCALAR

Scalar Quantities will have magnitude only



Car is travelling with a speed of 40 kmph

VECTOR

Vector Quantities will have both magnitude and direction



Car is travelling towards east with a speed of 40 kmph

SCALAR QUANTITIES

DISTANCE

The length of path traversed by an object.
Eg. 5kms, 10 km, 100 km

SPEED

The rate at which a distance is covered. Eg.
5 m/s, 10 m/s, 100 m/s

POWER

The rate of transfer of energy. Eg 10 watt,
100 Watt



VECTOR QUANTITIES

DISPLACEMENT

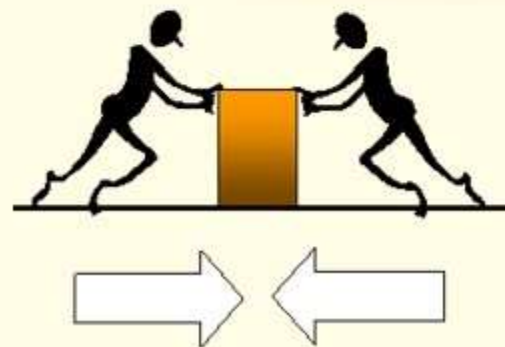
Shortest distance traversed by an object
Can be +ve, -ve and zero!
Eg. 5km - east, 10 km above

VELOCITY

Similar to Speed but has direction
Eg. 5 m/s up, 10 m/s in north

FORCE

- Always has a direction of action
- When we push a object it moves in that particular direction



SCALAR AND VECTOR QUANTITIES... THEIR DIFFERENCES

BASIS FOR COMPARISON	SCALAR QUANTITY	VECTOR QUANTITY
MEANING	Any physical quantity that does not include direction is known as scalar quantity.	Vector quantity is one, that has both magnitude and direction.
QUANTITIES	One-dimensional quantities means it is a simple number	Multi-dimensional quantities
CHANGE	It changes with the change in their magnitude.	It changes with the change in their direction or magnitude or both.
OPERATIONS	Follow ordinary rules of algebra.	Follow rules of vector algebra.
DIVISION	Scalar can divide another scalar.	Two vectors can never divide.



WHAT CHANGES HAPPEN WHEN WE DIVIDE A MATTER ???



TEMPERATURE
DENSITY
HARDNESS
ARE INTRINSIC
PROPERTIES

- 1 **Temperature** of a ice block will be same even if it cut into two pieces.
- 2 **Density** of the ice cubes will be same like that of the single piece
- 3 **The hardness** of ice as well the light passing capacity will not change when the ice block is broken into two





WHAT CHANGES HAPPEN WHEN WE DIVIDE A MATTER ???



MASS
VOLUME
FORCE ARE
EXTRINSIC
PROPERTIES

- 1 **Mass** of the ice block will not be same if it cut into two pieces.
- 2 **Volume** of the ice cubes will not be same like that of the single piece
- 3 The **Force** required to move the ice cubes will be different than the force required to move the ice block



POINTS OF DIFFERENCE BETWEEN INTRINSIC AND EXTRINSIC PROPERTIES

INTRINSIC PROPERTIES

These properties do not depend on

- The size of the material or
- The amount of the material in the system

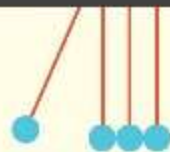
E.g. density, pressure conductivity temperature

EXTRINSIC PROPERTIES

These properties depend on

- The amount of matter present in the system

E.g. mass, volume force energy, electric charge



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